Estimating the effect of soil moisture on food production, farmer adaption and famine in Africa

Steve Berggreen[1*], Hannah Druckenmiller[2,3], Solomon Hsiang[1,3]

[1]Global Policy Laboratory, Stanford Doerr School of Sustainability, Stanford University, USA
[2]Division of Humanities and Social Sciences, California Institute of Technology, USA
[3]National Bureau of Economic Research, USA

[*]Corresponding author: stevebc@stanford.edu

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Abstract

Droughts are a major threat to food security, yet their contribution to large-scale historical famines is difficult to establish. The majority of post-war famines occurred in Africa, before the advent of satellite-based soil moisture observations. Here we reconstruct monthly surface soil moisture across Africa from 1945–2014 by combining historical weather observations with a machine-learning model informed by hydrological processes and soil constraints, trained on modern satellite data. We validate the reconstruction against independent soil moisture and vegetation observations and link growing-season soil moisture to production, yields and harvested area for four major staple crops. Climate impacts on agriculture are typically quantified through yields, which capture losses only on land that reaches harvest. We show that drought also substantially reduces harvested area, causing yield-based measures to understate the true scale of agricultural losses. A decline in soil moisture from 0.25 to 0.15 cm^3 cm^{-3} reduces total production by 33–48%, and for millet, the estimated production loss is three times as large as the yield loss. Production losses predicted by historical climate shocks closely track observed agricultural shortfalls, and famine probability rises in the year following a shock. The increase in famine risk following drought-related production losses is largest in low-income countries and autocracies. Our results shed new light on how drought translates into food-production collapse and show that historical assessments based on rainfall and yields alone can substantially understate the agricultural shocks that accompany famine.

Introduction

Drought is a major threat to food security[1], yet its contribution to large-scale historical famines remains difficult to establish. The majority of post-war famines occurred in Africa, where many of the most severe food crises unfolded before agricultural water availability could be observed from satellites[2]. Drought can reduce food production and rural incomes[1], while conflict, poverty and failures of public response are widely understood to shape whether those losses ultimately lead to famine[3,4]. Historical accounts document these forces acting together, often within the same crisis[2]. Distinguishing their roles requires both a measure of agricultural water stress that extends across a sufficient set of recorded famines, and accounting for how that stress changes the total quantity of food produced.

The first challenge is measuring agricultural drought itself. Because plants respond to soil water rather than rainfall alone, the same rainfall total can imply very different growing conditions depending on evaporative demand and local storage conditions. Satellite observations have transformed the measurement of soil moisture, but reliable coverage in Africa is largely limited to the satellite era, and evaluations of the ESA Climate Change Initiative record motivate particular caution before 1992[5,6]. The majority of modern-era African famines occurred before this observation window, which motivates the historical reconstruction (Fig. 1a).

The second challenge is measuring the agricultural consequences of drought. Most climate–agriculture assessments quantify weather impacts through crop yields, including nonlinear temperature effects and their implications for future agricultural productivity[7–9]. Recent research has also emphasized the role of water supply and the difficulty of representing crop water stress using precipitation alone[10–13]. Yet yield is output per harvested hectare and therefore conditions on cropland that successfully reaches harvest. If drought reduces the area planted, prevents planted crops from surviving to harvest, or otherwise reduces harvested area, part of the resulting food-production loss disappears from the yield denominator. For food security, however, the relevant quantity is total production, which depends jointly on how much each harvested hectare produces and how much land is ultimately harvested.

Existing evidence points to an important role for this extensive margin, but is concentrated in agricultural systems outside Africa[14–17]. Separating these margins can also distinguish agronomic responses from behavioural adjustments[18]. In sub-Saharan Africa, household surveys document stagnant crop yields and substantial losses associated with climate shocks, while recent work examines farmers' responses to shifting growing seasons[19–21]. These findings motivate measuring drought impacts on total production directly, and decomposing those effects into yield and harvested-area responses. This may be be especially important during severe food-security shocks.

To address these measurement gaps, we reconstruct monthly surface soil moisture across

Africa for 1945–2014. We combine a long historical rainfall compilation[22] with temperature[23], soil[24] and topographic information in a machine-learning model trained on modern SMAP Level-4 surface soil moisture[25]. The model is informed by hydrological processes and soil-storage constraints, incorporating antecedent moisture and weather histories, soil and landscape characteristics, and bounds based on local storage capacity. The resulting approach extends machine-learning soil-moisture estimation back into the pre-satellite era and complements land-data assimilation systems developed for African drought monitoring[26,27].

We then use this reconstruction to estimate crop-specific dose–response functions linking growing-season temperature and soil moisture to production, yield and harvested area for cassava, maize, millet and sorghum, which together account for more than half of calories produced in sub-Saharan Africa, over 1961–2014[28]. We use these responses to construct climate-predicted caloric production losses, compare them with observed agricultural shortfalls, and examine their timing relative to documented famine years. Finally, we test whether the relationship between drought-related production losses and famine differs with income, political institutions, conflict and access to the coast.

Reconstructing Africa's historical soil moisture

We reconstruct monthly surface soil moisture across Africa from 1945 to 2014, extending coverage back nearly five decades before the external satellite-validation period used here (Fig. 1a). The reconstruction is driven by a long historical rainfall station record[22], together with temperature and spatial information on soils and landscape characteristics. Modern meteorological data and SMAP Level-4 soil moisture provide the training environment for learning how these inputs map into surface moisture conditions[22,29]. Because soil moisture is a state variable—a stock that depends on prior moisture conditions as well as current weather—the model incorporates antecedent weather and moisture histories and evolves moisture recursively through time, while bounding predictions according to local storage capacity (Methods).

Figure 1b first evaluates the model in the modern period using withheld SMAP observations. The final 24 months are excluded from training, with an additional 12-month gap separating the training and test periods (Methods). In this holdout sample, predicted month-to-month changes in soil moisture explain 76% of the observed variation, with a root-mean-square error of 0.015 $cm^3\ cm^{-3}$. This exercise tests short-run predictive performance when the preceding moisture state is observed. This is different from the historical reconstruction, which evolves recursively using its own previous predictions.

The historical series combines these short-run updates with a complementary model of moisture levels, which anchors the recursive trajectory and limits accumulated drift (Methods). Figure 1c illustrates the resulting reconstruction during two contrasting Sahelian years. The wet conditions of 1966 and the widespread dryness of 1973 emerge as sharply different

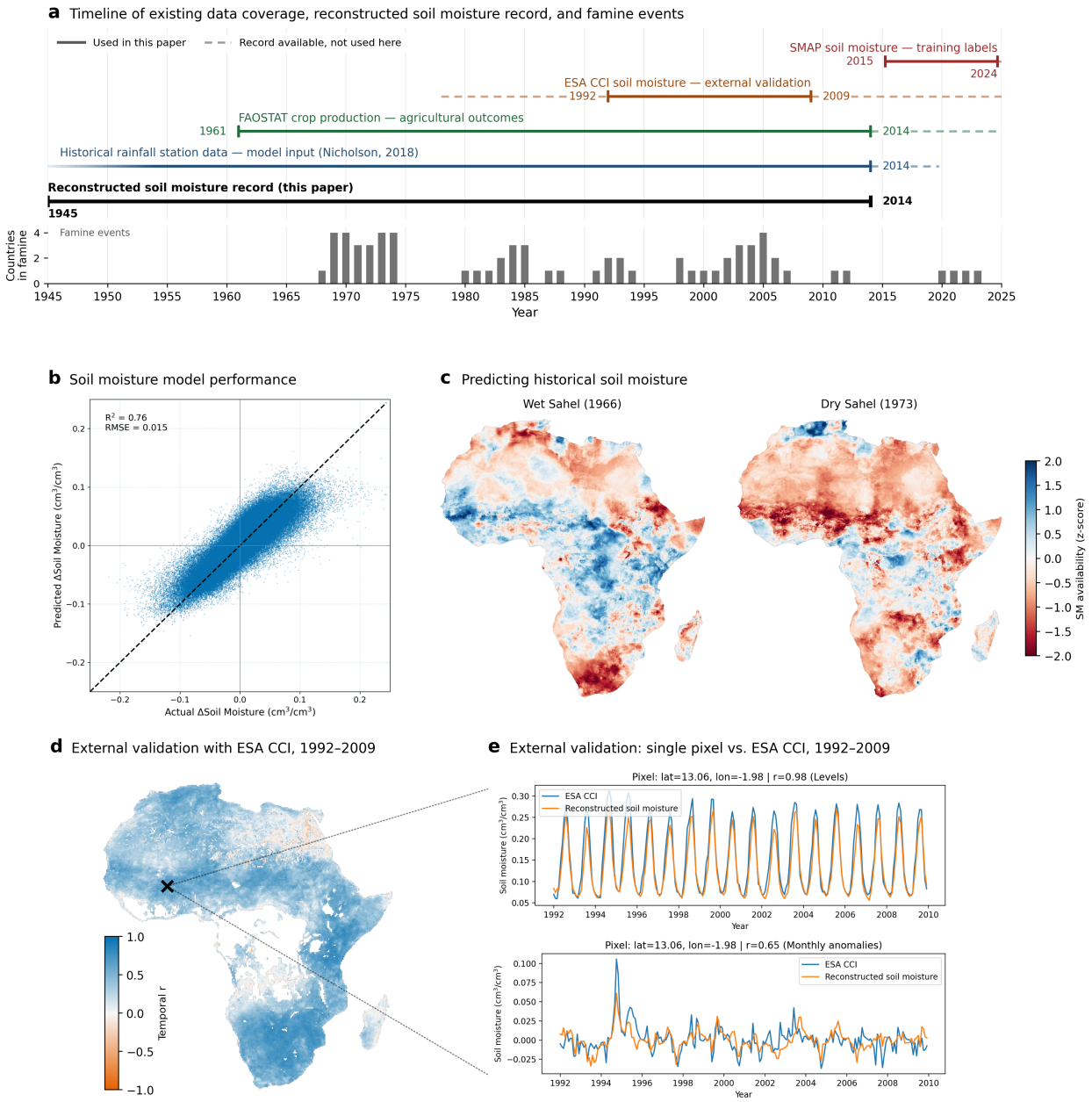


Figure 1: Reconstructing historical soil moisture across Africa. **a**, Coverage of the reconstruction, input and comparison datasets, and coded famine years. Dashed intervals indicate available records outside the windows used. **b**, Predicted versus observed monthly soil moisture changes in the withheld modern SMAP L4 test split ($R^2 = 0.76$; RMSE = 0.015 cm^3 cm^{-3}). **c**, Standardized reconstructed moisture availability across Africa in 1966 and 1973. **d**, Pixel-wise temporal correlation with ESA CCI during 1992–2009 after removing grid-cell-specific monthly climatologies. Missing data in ESA CCI due to dense vegetation cover is masked out. **e**, Levels and deseasonalized anomalies at a location in Sahel.

continent-wide moisture states. The 1973 pattern coincides with famine years in Mauritania, Mali, Niger and Ethiopia in the event coding used here, illustrating the type of historical

agricultural exposure that cannot be observed directly from the modern satellite record.

We next assess whether the free-running historical reconstruction captures independently observed variation in soil moisture. Figure 1d compares the reconstructed series with ESA CCI over 1992–2009, after removing each grid cell's calendar-month climatology. This deseasonalization is important because two products can agree on the timing of wet and dry seasons while disagreeing on which particular years were unusually dry or wet. The map therefore evaluates whether the reconstruction captures departures from the local seasonal cycle, rather than simply reproducing climatology.

Agreement in these anomalies is broadly positive across the continent, with the strongest performance in several famine-prone regions, including much of the Sahel, and weaker performance in very arid and humid environments such as Egypt and the Democratic Republic of the Congo (Fig. 1d). Across all valid grid cells, the mean [median] anomaly correlation is 0.38 [0.40] (Methods). Figure 1e illustrates the comparison at a Sahelian location, where the correlation with ESA CCI is 0.98 in levels and 0.65 after removing the seasonal cycle.

Drought compounds production losses through yields and harvested area

We next estimate how growing-season soil moisture affects production for cassava, maize, millet and sorghum, four staple crops that together account for more than half of crop calories produced in sub-Saharan Africa (Methods). We use crop-specific growing seasons[30] and spatial suitability weights[31] to weight climate exposure across the locations and months most relevant for each crop. We estimate soil-moisture responses jointly with growing-season temperature, given the well-documented nonlinear effects of heat on crop yields[7–9]. Production, yield and harvested area are estimated under the same specification (Methods), and Fig. 2 shows the fitted soil-moisture response relative to 0.25 $cm^3\ cm^{-3}$, holding temperature constant.

Lower soil moisture is associated with substantial declines in total production across all four crops (Fig. 2a). Moving from 0.25 to 0.15 $cm^3\ cm^{-3}$ reduces fitted production by roughly one-third to one-half, depending on the crop. The exposure distributions also differ across crops in expected ways, with cassava being concentrated in relatively wetter conditions, while millet and sorghum are more heavily represented at lower soil-moisture levels (Fig. 2a).

Yield responses capture only part of these production losses. Across the same moisture contrast, fitted yield declines range from about 11% to 36%, while harvested-area declines range from about 19% to 26%. The relative importance of the two margins differs across crops. Maize and cassava exhibit larger yield responses, whereas harvested-area losses are especially important for millet and sorghum. For millet, the contrast is particularly stark: the fitted production loss is about three times as large as the yield loss.

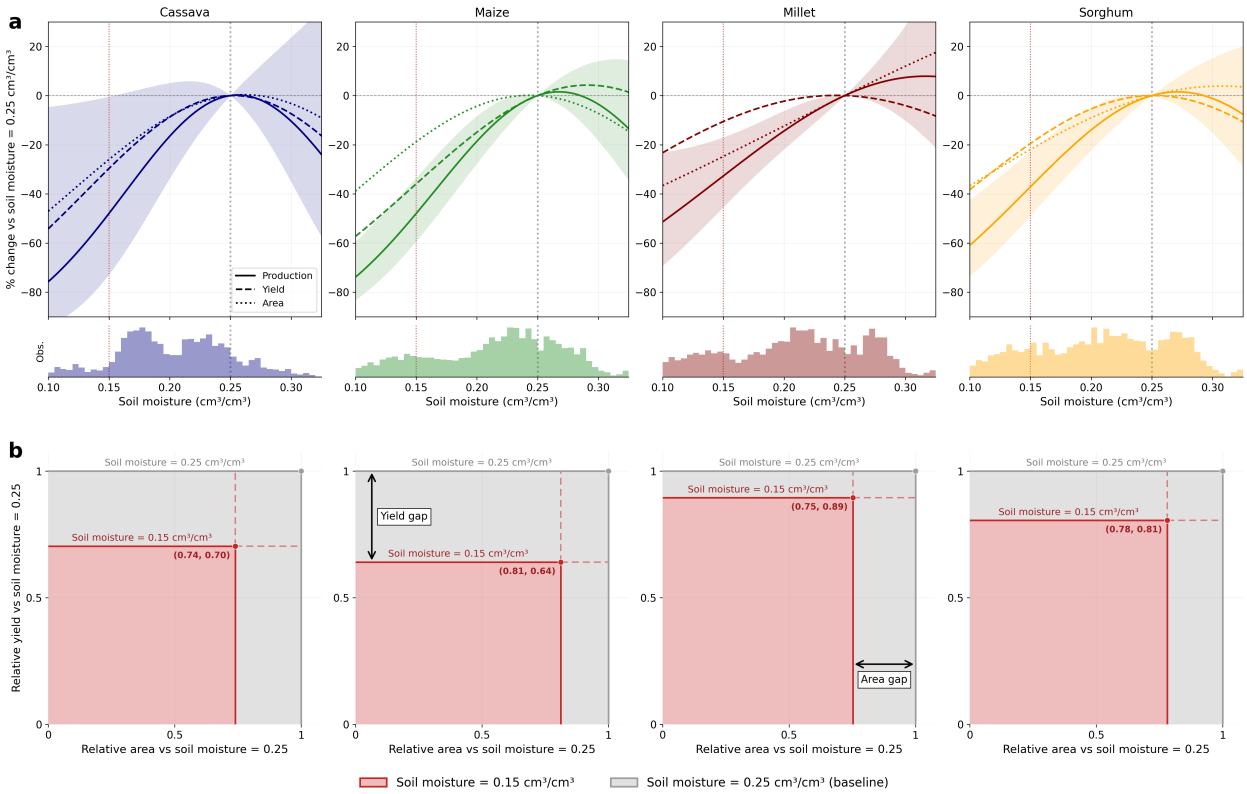


Figure 2: Harvested-area losses amplify the agricultural response to dry soils. a, Crop-specific production, yield and harvested-area responses to soil moisture in quadratic temperature–soil moisture models. Responses are normalized to 0.25 cm^3 cm^{-3}; shading shows 95% confidence intervals for production. Histograms show moisture support in the plotting sample. Regressions include country–crop fixed effects and country trends, with country–crop-clustered standard errors. **b**, Yield and area ratios at 0.15 relative to 0.25 cm^3 cm^{-3}. Rectangle width is relative harvested area, height is relative yield and their product is relative production; labels give (relative area, relative yield) at 0.15 cm^3 cm^{-3}.

Figure 2b shows how these concurrent responses translate into production losses. Because total production is the product of yield and harvested area, production at soil moisture s relative to the baseline $s_0 = 0.25$ cm^3 cm^{-3} is the product of relative yield and relative harvested area, $P(s)/P(s_0) = [Y(s)/Y(s_0)] \times [A(s)/A(s_0)]$, so losses along the two margins compound. At $s = 0.15$ cm^3 cm^{-3}, relative yield times relative area is $0.70 \times 0.74 = 0.52$ for cassava, $0.64 \times 0.81 = 0.52$ for maize, $0.89 \times 0.75 = 0.67$ for millet and $0.81 \times 0.78 = 0.63$ for sorghum, corresponding to production losses of 48%, 48%, 33% and 37%. For maize, the decline is driven more strongly by yield, with harvested area providing an additional reduction in output. For millet, by contrast, yield changes relatively little while harvested area contracts substantially, so a yield-only measure misses most of the production response. At the 0.25 to 0.15 cm^3 cm^{-3} contrast, relying on yields alone would understate the fitted production decline by about 12 percentage points for maize and 22 percentage points for

millet.

Harvested area combines several underlying responses that cannot be separated in the national data, including decisions not to plant and the failure of planted crops to reach harvest. From a food-availability perspective, however, both mechanisms have the same implication: land that is never planted or never reaches harvest reduces total production while disappearing from the denominator of conventional yield measures. This illustrates why yield-based assessments can substantially understate drought-related food-production losses. The result extends earlier evidence on area responses[14,17,20] to a continent-wide historical setting in which agricultural water stress and food insecurity can be studied jointly.

Production shocks precede famine and capture crisis severity

We next translate the crop-specific climate responses into country-level caloric production losses. For each crop and year, we compare predicted production under observed growing-season climate with production under the country's long-run crop-specific climate conditions, and aggregate across crops using caloric conversion factors (Methods). The resulting measure thus isolates the component of production loss predicted by temperature and soil moisture.

As a first validation, climate-predicted losses closely track observed agricultural shortfalls in the broader country–year panel (Fig. 3a). A one-percentage-point increase in predicted loss is associated with a 0.92-percentage-point increase in observed caloric loss in the same year (95% confidence interval, 0.63–1.20). The response is concentrated in the shock year and subsequently reverses, consistent with temporary production shortfalls rather than persistent differences across countries.

Famine risk, however, responds with a lag (Fig. 3b). A ten-percentage-point predicted production loss is associated with a 0.54-percentage-point increase in famine probability in the same year and a 0.84-percentage-point increase one year later (95% confidence interval, 0.06–1.63 for the latter). There is little evidence of an increase before the shock. Combining current and previous-year losses, a ten-percentage-point increase in cumulative predicted caloric loss is associated with a 0.97-percentage-point increase in famine probability, equivalent to a 36% increase relative to the sample famine rate of 2.7% (Methods). Shifting this exposure window two years backward or forward produces estimates close to zero (Fig. 5).

This temporal sequence is consistent with agricultural shocks propagating into food crises with a delay. In the Sahel, for example, harvests tend to occur late in the calendar year[30], so losses recorded in year t may affect food availability, prices and household resources most strongly in year $t + 1$. The estimates do not identify which of these channels drives the lag, and famine is coded throughout documented event spells rather than only at onset (Methods).

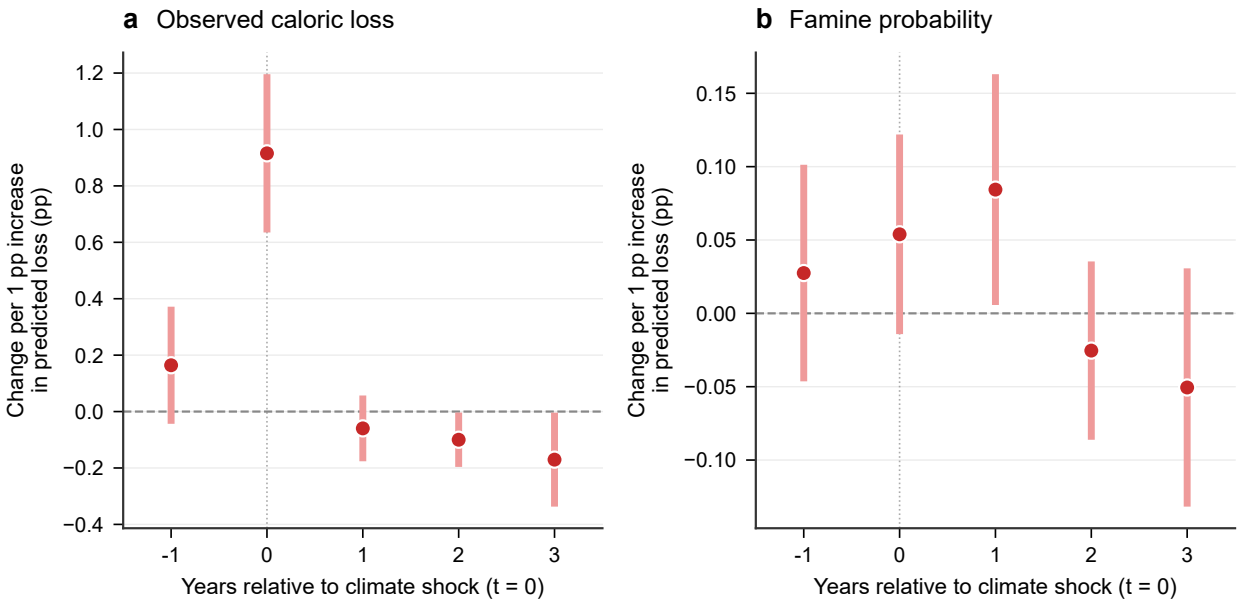


Figure 3: Agricultural losses coincide with climate shocks and famine risk rises subsequently. Distributed-lag coefficients for observed caloric production loss (**a**) and famine-year probability (**b**), per one-percentage-point increase in predicted production loss by the temperature-soil moisture model. The plotted window includes one year before the shock, the shock year and three subsequent years. Shading denotes 95% confidence intervals.

We next ask whether the same framework better captures the magnitude of agricultural losses during historical famine episodes. We compare the temperature–soil moisture production model with a conventional temperature–precipitation model estimated on crop yields[7,9,32]. This comparison incorporates both advances developed above: soil moisture rather than precipitation as the measure of agricultural water stress, and total production rather than yield as the measure of agricultural loss.

Among famine country–years with observed below-trend production, and using predictions from models estimated without the focal country–year, the soil-moisture production model captures substantially more of the magnitude of observed shortfalls (Fig. 4). For each additional one-percentage-point observed production loss, the model predicts an additional 0.51 percentage points of loss, compared with only 0.21 percentage points for the temperature–precipitation yield model. Root-mean-square error also falls from 23.4 to 19.7 percentage points. The soil-moisture production model therefore recovers substantially more of the severity of famine-year agricultural losses than the conventional precipitation–yield specification.

This difference in model performance is especially visible during major historical crises. Somalia's 2011 famine followed consecutive failed rainy seasons and exceptionally poor harvests, but unfolded amid protracted conflict, rapidly rising food prices and severe constraints on humanitarian access[33]. Observed caloric production fell by 58.9%, and while the soil-

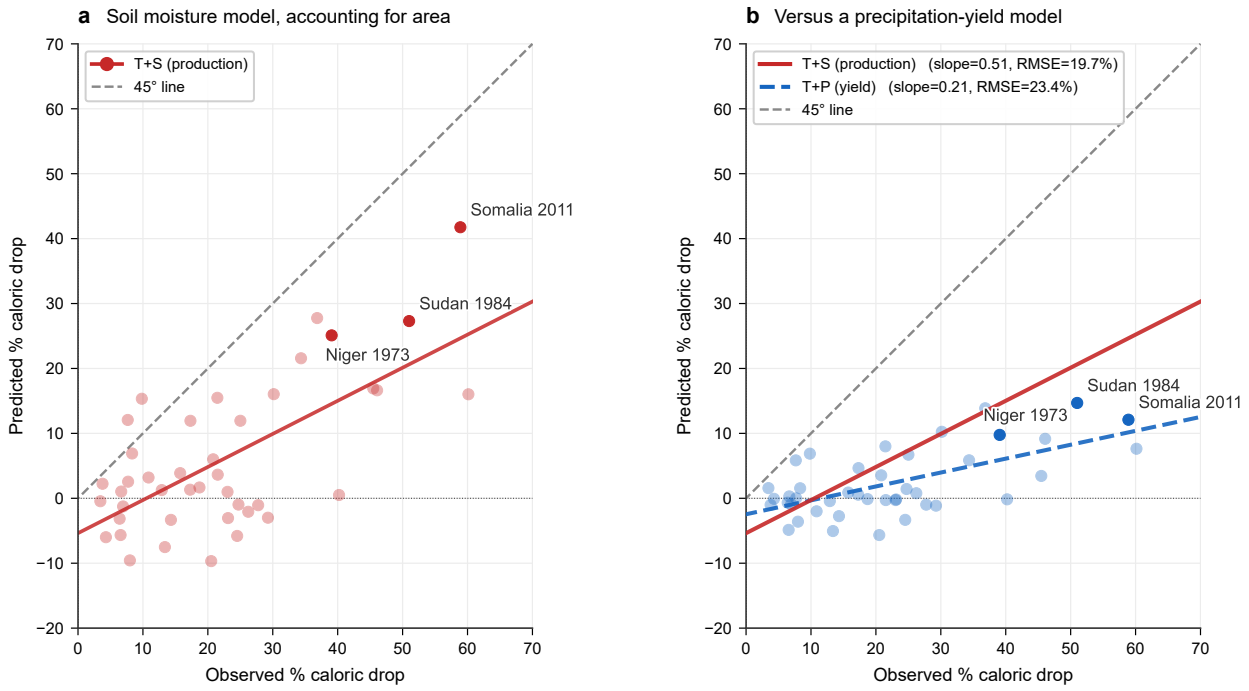


Figure 4: Soil moisture and harvested area improve the prediction of famine-year shortfalls. Observed and out-of-sample predicted caloric production losses in the 42 famine country-years with observed caloric shortfalls. Each prediction comes from models estimated without that country-year. **a**, Temperature–soil (T+S) moisture production predictions versus a one-for-one reference line. **b**, Comparison with a temperature–precipitation (T+P) yield model.

moisture production model predicts a 41.8% loss, the precipitation–yield model predicts only a 12.1% loss (Fig. 4). Niger's 1973 crisis occurred at the culmination of the prolonged 1968–1973 Sahel drought, when crop failures, livestock losses and rising cereal prices contributed to severe food shortages across the region[34]. Here the observed loss and corresponding predicted losses are 39.1%, 25.1% and 9.8% (Fig. 4). These cases illustrate how accounting for both soil moisture and harvested-area contraction improves the prediction of disrupted food production.

However, a climate-only model should not be expected to reproduce the full magnitude of famine-year production collapses, since conflict, market disruption, displacement and failures of public response can independently depress agricultural output and amplify food insecurity[3,4]. The unexplained component in Fig. 4 thus points to the role of broader economic and political forces that shape severe food crises, which motivates the follow-up question: do similar climate-induced production shocks translate into different famine risks across different institutional and economic settings?

Famine risk varies with economic and institutional conditions

The same climate-induced production loss need not translate into the same risk of famine. Economic resources can support food imports, household consumption and public relief; political institutions can affect the incentives and constraints facing governments as shortages emerge; and conflict can disrupt production, markets and humanitarian access. Historical famine scholarship has long emphasized these political and distributional mechanisms[2,3,35]. We therefore examine whether the relationship between predicted agricultural losses and famine varies with income, political institutions, conflict and geographic isolation.

We interact the sum of predicted losses in the focal and preceding year with each moderator in turn, using income, regime type and political institutions from V-Dem[36], together with measures of conflict and geographic isolation (Methods). We control for country–decade fixed effects and cluster standard errors by country (Fig. 5). Identification therefore comes from year-to-year variation in climate-induced production losses and famine incidence within the same country and decade, rather than from persistent differences across countries or long-run changes over time.

The estimated famine response is largest in lower-income countries and autocracies. Using V-Dem's regime classification, a ten-percentage-point increase in cumulative loss is associated with a 1.22-percentage-point increase in famine probability among autocracies (95% confidence interval, 0.23–2.22). Among lower-income countries, the corresponding increase is 1.47 percentage points (0.32–2.62). By contrast, the middle/high-income group contains *no* recorded famine years and the democracy group only one, so their near-zero slopes should not be interpreted as evidence that income or democracy eliminates famine risk.

Since most country-year observations are classified as autocracies, the binary regime distinction masks potentially important variation in political institutions among autocracies. We therefore also use V-Dem's Liberal Democracy Index, a continuous measure combining electoral accountability with constraints on political power such as civil liberties, rule of law, judicial independence and checks on the executive[36]. This measure allows us to capture institutional variation even among countries that remain classified as autocracies. A ten-percentage-point increase in cumulative production loss is associated with increases in famine probability of 0.7-2.0 percentage points in the lowest three quartiles of the index, compared with essentially zero in the highest quartile. While the pattern is not monotonic across the lower three quartiles, the sharp decline at the top of the distribution is consistent with the view that greater electoral accountability and stronger constraints on political power can reduce famine risk[3].

Conflict produces the largest point estimates. A ten-percentage-point production loss is associated with a 3.26-percentage-point increase in famine probability during war, compared with 1.13 percentage points during minor conflict and 0.77 in country-years without recorded

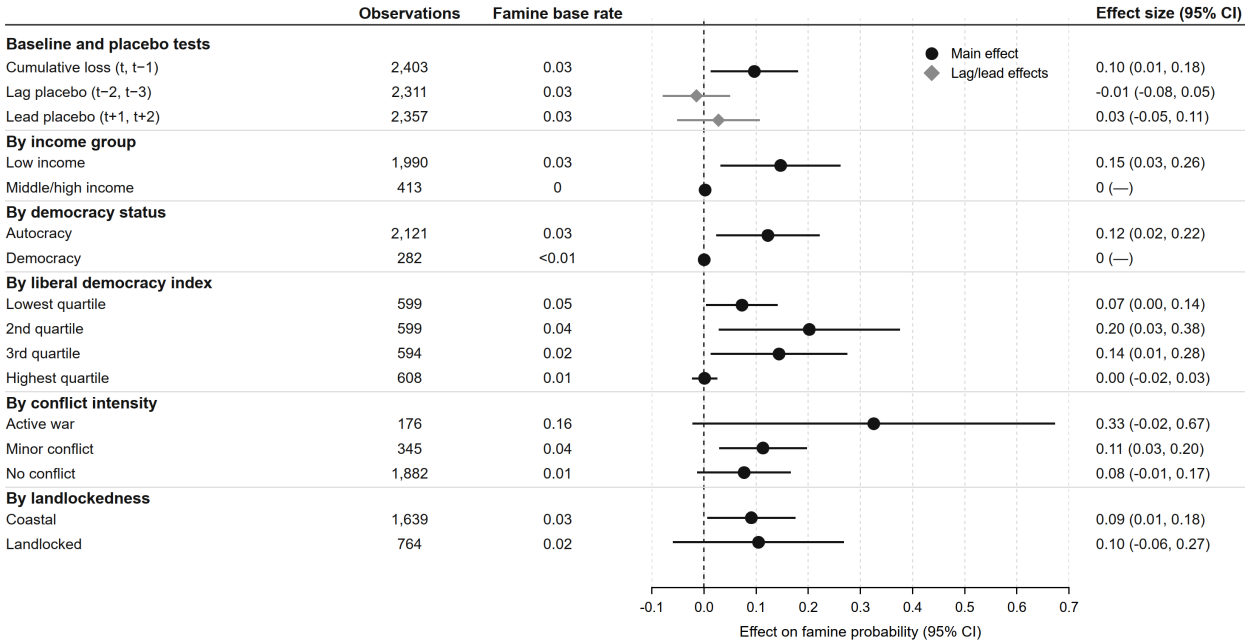


Figure 5: The loss–famine relationship varies across economic and political settings. Baseline and displaced-window estimates represent separate country–decade fixed-effects regressions. Heterogeneity rows use interaction models, with subgroup slopes recovered by linear combinations and standard errors clustered by country. Exposure is the sum of current and previous annual losses, in fractional units. Multiplying a reported slope by 10 gives the percentage-point probability change for a ten-percentage-point increase in summed loss. Columns report subgroup observations, famine base rates and 95% confidence intervals.

conflict. The war estimate is very imprecise, however, with a 95% confidence interval spanning −0.22 to 6.73 percentage points.

As a proxy for whether greater access to trade and external food assistance may alleviate domestic production shocks, we compare coastal and landlocked countries. The estimated responses are broadly similar, however, with substantial uncertainty for the landlocked group.

These results suggest that the consequences of climate-induced agricultural losses depend on the social and political environment in which they occur. They do not identify the causal effects of higher income, democratization or conflict itself. Instead, we exploit year-to-year variation in soil-moisture conditions that is plausibly exogenous to contemporaneous economic and political conditions, providing a common environmental shock against which differences in vulnerability can be compared. Overall, the results are consistent with the broader view that severe food crises emerge from the interaction of environmental shocks with economic resources, political institutions and conflict.

Discussion

Our results help reconcile two views of famine that are often considered separately. Droughts can generate large shocks to domestic food production, but the same agricultural shock does not typically produce the same humanitarian consequences across societies. By reconstructing soil moisture before the satellite era, we show that drought-related production losses accompanying historical African famines were both substantial and systematically understated by conventional measures based on rainfall and crop yields. At the same time, the relationship between these production losses and famine varies with the economic and political conditions under which they occur. Environmental shocks therefore help determine the size of the food-production shortfall, while institutions, resources and conflict shape how strongly that shortfall translates into famine[2,3,35].

A central reason agricultural losses have been understated is that crop yield conditions on land that reaches harvest. Droughts reduce output on land that is harvested (yield losses), but we show that the effects on harvested land itself are meaningful and comparable in magnitude to yield losses. Across the four staple crops studied here, this extensive margin substantially amplifies the estimated response to dry soils. For millet in particular, the estimated total production decline is approximately three times the yield decline during severe soil moisture deficits. Climate–agriculture assessments concerned with food security should therefore account for both of these dimensions.

These findings inform how we understand the causes of historical famines. Drought-related production losses appear to be an important part of severe food crises, but they are not sufficient to explain them. Historical work has long emphasized entitlement failures, political accountability, conflict and the distribution of food as central determinants of famine[2,3,35]. Our results are consistent with this view: environmental shocks shape the size of the agricultural shortfall, while economic and political conditions shape how strongly that shortfall translates into famine.

A few considerations are important when interpreting these results. Our analysis focuses on four major staple crops and therefore does not capture the full food system, including other crops, livestock, and trade. FAO crop data also cannot distinguish whether reductions in harvested area reflect lower planting, crop failure before harvest or other adjustments[28]. The heterogeneity analysis is similarly descriptive rather than causal. Differences across income levels, political institutions and conflict settings show where comparable climate-induced production losses are more or less strongly associated with famine, but they do not identify the causal effects of democratization, higher income or conflict itself.

More broadly, the results suggest that assessments of climate-related food insecurity should move beyond yield responses alone and account for the full effect of water stress on production[32,37,38]. Combining reconstructed soil moisture with seasonal forecasts and climate change projections may also help identify emerging agricultural stress earlier.

The policy implications are correspondingly broader than drought monitoring alone. Better measurement of climate-induced production losses can help distinguish places facing large agricultural shocks from places where social, economic or political conditions make those shocks especially dangerous. More generally, understanding famine risk requires measuring both the severity of the environmental shock and the conditions that determine whether societies can absorb it.

Methods

Study design

We reconstruct monthly surface soil moisture across Africa for 1945–2014 and link crop-season exposures to national agricultural outcomes over 1961–2014. SMAP observations provide training labels for the reconstruction, and ESA CCI soil moisture provides an external comparison over 1992–2009. The analysis proceeds in five stages that correspond to the main figures: reconstruction and validation; crop-specific yield and area responses; the timing of agricultural losses and famine; prediction of famine-year production shortfalls; and heterogeneity by income, institutions and conflict.

Soil moisture and meteorological data

SMAP training labels. We train on monthly composites of volumetric surface soil moisture (0–5 cm) from the NASA SMAP Level-4 (L4) product on its nominal 9-km grid[25]. L4 assimilates SMAP brightness temperatures into a meteorologically driven land-surface model and provides three-hourly estimates, including between satellite overpasses. This continuity allows us to construct monthly training targets and to learn antecedent-moisture dynamics. We chose L4 over the enhanced L3 radiometer product, whose coverage depends on observation availability and retrieval conditions[39]. In an evaluation against 362 soil moisture stations in the United States, L4 performed best among the products tested, including the L3 radiometer product, and showed the strongest agreement over croplands, with a Spearman correlation of 0.60[40].

Precipitation. Modern model inputs are monthly MSWEP precipitation[29]. Historical inputs come from the African rainfall compilation of Nicholson and colleagues, which extends station observations back well before the satellite period[22]. We interpolate station totals to a regular grid by inverse-distance weighting (power 2) with an adaptive search radius that expands until at least four stations are found, following precedent in gridded African rainfall reconstruction[41]. Monthly precipitation rasters are aligned to the soil moisture grid by bilinear interpolation.

Temperature. Temperature predictors are Berkeley Earth monthly mean, maximum and minimum temperature, with absolute temperatures reconstructed by adding anomalies to the corresponding monthly climatology[23]. Diurnal range is maximum minus minimum temperature. Mean temperature, diurnal range, precipitation and a seasonal radiation proxy enter contemporaneously and at six monthly lags. The radiation proxy is the product of mean temperature, the cosine of latitude and a sinusoidal seasonal term, and serves as an empirical predictor of seasonal energy conditions. Dates are intersected across weather and

label files before features are built, so that each lagged observation refers to the preceding calendar month.

Soils and landscape. Static predictors are SoilGrids sand, clay, bulk density, coarse-fragment volume and organic carbon at 5–15 cm depth[24]; elevation and slope derived from SRTM[42]; mean river discharge over 1984–2022 from GloFAS[43], transformed as $\log(1+Q)$; and Köppen–Geiger climate zones[44]. The ten most frequent climate zones enter as separate indicators and the remaining zones are pooled. These static fields are held fixed in the historical reconstruction.

Storage constraint. Bulk density is converted from the stored SoilGrids units to g cm^{-3}. Porosity is $\phi_g = 1 - \rho_{b,g}/2.65$, truncated to $[0.30, 0.65]$, and the working moisture ceiling is $c_g = 0.90\phi_g$. Training labels are capped at this ceiling, and reconstructed moisture is bounded between 10^{-3} and c_g in volumetric units.

Machine-learning reconstruction

Change and level models. Let S_{gt} denote surface moisture in grid cell g and month t, and let X_{gt} contain the static predictors and the current and six preceding months of weather features. A first random forest, the change model, predicts $\Delta S_{gt} = S_{gt} - S_{g,t-1}$ from X_{gt}, the preceding moisture state and the relative fullness $S_{g,t-1}/c_g$, which is clipped to $[0, 1.2]$ in training. A second forest, the level model, predicts S_{gt} directly from X_{gt} without lagged soil moisture.

Each forest is a scikit-learn RandomForestRegressor with 50 trees and random seed 42; other hyperparameters take library defaults. We train the change model on a random 5% sample of pixels drawn without replacement (seed 42) and the level model on a 1% sample (seed 43). Sampled pixels contribute all available months, and rows with missing features or labels are excluded. The final 24 months are held out for testing, and the 12 months before them are excluded from training.

Modern performance. We evaluate the change model on the test split using $R^2 = 1 - \sum_j (y_j - \hat{y}_j)^2 / \sum_j (y_j - \bar{y})^2$ and RMSE $= \sqrt{n^{-1} \sum_j (y_j - \hat{y}_j)^2}$, obtaining $R^2 = 0.76$ and RMSE $= 0.015$ cm^3 cm^{-3} (Fig. 1b). In the test period the lagged state is the observed SMAP moisture, so this evaluation measures one-step prediction of monthly change.

Fusion and recursive prediction. In the historical reconstruction, the previous fused prediction supplies the lagged state. We first form

$$\widetilde{S}_{gt} = \mathcal{C}_g \left[\widehat{S}_{g,t-1} + f_\Delta(X_{gt}, \widehat{S}_{g,t-1}, \widehat{S}_{g,t-1}/c_g) \right], \tag{1}$$

$$\widehat{S}_{gt} = \mathcal{C}_g \left[(1-w)\widetilde{S}_{gt} + w f_L(X_{gt}) \right], \qquad w = 0.15, \tag{2}$$

where f_Δ and f_L are the change and level models and $\mathcal{C}_g$ clips predictions to the moisture bounds. The level prediction receives 15% weight and the recursive update 85%, which anchors the trajectory and limits accumulated drift while retaining moisture memory. Positive predicted changes are set to zero when relative fullness reaches 0.98, and relative fullness is bounded at one at inference.

The recursion starts in January 1945 with each cell initialized at $\min(0.15, 0.95c_g)$. A cell's state is updated whenever its full lag history is available and otherwise retains its previous value, so the earliest months carry the initial value.

Historical maps. Figure 1c shows standardized annual moisture availability for 1966 and 1973. Monthly availability is $\min\{\max(\widehat{S}_{gt}, 0.05), 0.25\} - 0.05$ in $\mathrm{cm}^3\ \mathrm{cm}^{-3}$, which counts moisture between 0.05 and 0.25 $\mathrm{cm}^3\ \mathrm{cm}^{-3}$. We sum it over the twelve months of each year and standardize each grid cell by its mean and standard deviation across years.

External validation with ESA CCI

We use monthly C3S-distributed ESA CCI combined active–passive soil moisture (release v202505) over January 1992–December 2009, giving up to 216 monthly observations per cell[5]. The 1992 start follows evidence of limited earlier CCI coverage in East Africa[6]. We intersect the available calendar months of the two products, reproject CCI onto the reconstruction grid by nearest-neighbour resampling and mask both products to African country geometries.

For each pixel we compute the temporal Pearson correlation over finite pairs, requiring at least 12 overlapping months. The raw correlation is $r_g = \mathrm{corr}_t(S_{gt}^{\mathrm{CCI}}, \widehat{S}_{gt})$. To remove the seasonal cycle, we demean each product by grid cell and calendar month over the validation period:

$$A_{gt}^{(q)} = S_{gt}^{(q)} - \overline{S}_{g,m(t)}^{(q)}, \qquad r_g^{\mathrm{anom}} = \mathrm{corr}_t(A_{gt}^{\mathrm{CCI}}, A_{gt}^{\mathrm{RF}}). \tag{3}$$

Figure 1d maps r_g^{anom}, and its mean and median across valid grid cells are reported in the main text. Figure 1e shows a location at approximately 13.06°N, 1.98°W, drawn at random from the Sahel.

We further compute pooled spatiotemporal RMSE over finite paired grid-cell-month observations; RMSE over a non-arid mask (CCI temporal mean above 0.05 and temporal standard deviation above 0.01 $\mathrm{cm}^3\ \mathrm{cm}^{-3}$); unbiased RMSE after subtracting each cell's full-period temporal mean from both products; and a pooled predictive R^2 from complete paired

time series at up to 1,000 randomly selected pixels (seed 123).

Agricultural outcomes and crop-season exposures

Production, yield and area. FAOSTAT provides production (tonnes), yield and harvested area (hectares) for cassava, maize, millet and sorghum over 1961–2014[28]; production equals harvested area times yield after unit conversion. We log positive outcomes for estimation and exclude zero or missing values from the corresponding regression.

Calendars and spatial weights. Rainfed growing seasons follow MIRCA2000 cell-specific calendars, with multiple sub-crop entries weighted by their reported cropped areas[30]. The four rainfed crop codes are 28, 32, 33 and 37. For seasons crossing December, months from the start through December belong to the preceding year and months through harvest belong to the harvest year. Across cells, exposures are weighted by GAEZ rainfed potential-yield layers, using the pixel-wise maximum of pearl and foxtail millet suitability for millet[31]. We use the 1999 CShapes country layer, harmonize country names and construct one observation per country, crop and harvest year. Static calendars and suitability layers are applied to the full historical panel. We retain rows with total suitability weight of at least 10^{-9} and, before spatial aggregation, drop MIRCA rows that lack any listed climate variable, cropped area or suitability weight, so that all climate moments share a common support. Within-season raster means use available values.

Exposure moments. Temperature regressors are growing-season averages of power moments of daily maximum temperature, first aggregated to monthly rasters. Soil moisture and precipitation regressors are monthly power moments. We average these across seasons within a cell using MIRCA area, and then across cells using GAEZ suitability. The soil moisture second moment is therefore the average of monthly squared moisture, $\langle S^2 \rangle$, and the same construction applies to temperature. The preferred quadratic temperature–soil moisture (T+S) model uses the first two moments of each variable. The comparison temperature–precipitation (T+P) model replaces soil moisture with precipitation.

Response model. For crop k, country i and year t, we estimate

$$\log Z_{ikt} = \beta_{1k}\langle T \rangle_{ikt} + \beta_{2k}\langle T^2 \rangle_{ikt} + \delta_{1k}\langle S \rangle_{ikt} + \delta_{2k}\langle S^2 \rangle_{ikt} + \alpha_{ik} + \gamma_i t + \epsilon_{ikt}, \tag{4}$$

where Z is production, yield or harvested area. Regressions are fitted separately by crop and outcome, absorb country–crop fixed effects and country-specific linear trends, and cluster standard errors by country–crop panel (equivalent to country within a single crop). They include no common year fixed effects. This fixed-effects approach follows the climate-

econometrics literature[45,46] and identifies weather responses from within-country fluctuations that are uncorrelated with other determinants of outcomes.

Figure 2 evaluates the fitted moisture polynomial at s and $s_0 = 0.25$ and plots $100\{\exp[\delta_{1k}(s - s_0) + \delta_{2k}(s^2 - s_0^2)] - 1\}$, holding temperature constant. Temperature curves use 32°C as the reference. We obtain uncertainty for the coefficient contrast with Stata's nlcom and exponentiate the interval endpoints to percentage changes. The figure shows confidence bands for production and point estimates for yield and area. The rectangles in Fig. 2b multiply the fitted yield and harvested-area ratios.

Interpretation of area. Writing production as $P = Y A_h = Y A_p \theta$, where A_p is planted area and $\theta = A_h/A_p$, gives $\partial \log P/\partial C = \partial \log Y/\partial C + \partial \log A_p/\partial C + \partial \log \theta/\partial C$. FAOSTAT harvested area combines the planting and survival components. A yield-only analysis understates a production loss when area contracts and overstates it when area expands[15,16,18].

Predicted and observed caloric production losses

Predicted losses. We convert production to proportional caloric totals using factors of 1.60 for cassava, 3.65 for maize, 3.78 for millet and 3.39 for sorghum; a common mass-to-energy conversion cancels from the loss ratios. Let $C_{ikt} = a_k P_{ikt}$ and let $f_k(X_{ikt})$ denote the estimated log-production climate component. Replacing every exposure moment with its country–crop long-run mean gives $\bar{X}_{ik}$ and counterfactual calories

$$C_{ikt}^{\mathrm{cf}} = C_{ikt} \exp\{f_k(\bar{X}_{ik}) - f_k(X_{ikt})\}, \qquad L_{it} = 1 - \frac{\sum_k C_{ikt}}{\sum_k C_{ikt}^{\mathrm{cf}}}. \tag{5}$$

Long-run means are taken over the moment variables themselves, including mean squared exposures. The production and yield variants apply their respective climate functions. Figure 4 expresses L in percent, and the famine regressions use L as a fraction, so that $L = 0.10$ denotes a 10% loss. Negative values are retained and represent predicted gains relative to the counterfactual. Crop-specific counterfactual changes are weighted by observed crop production. For the country-year panel used in Figs. 3 and 5, we fit the crop-response models on all retained years after removing upper-tail observations whose detrended log-production residual exceeds three within-panel standard deviations, once at least ten non-missing residuals have accumulated in the panel up to that observation. The lower tail is retained. For the famine-year comparison in Fig. 4, predictions are out of sample: for each focal famine country-year, we re-estimate the crop-response models of each specification excluding that country-year and use the resulting coefficients to compute its predicted loss.

Observed losses. Actual calories are summed across the included crops. We regress log aggregate calories on country intercepts and country-specific linear time trends. With $\widehat{m}_{it}$

the fitted log-trend production, observed loss is $O_{it} = 1 - \exp(\log C_{it} - \widehat{m}_{it})$. Figure 4 includes the famine country-years with $100O_{it} \geq 0$ (42 of 65). Its comparison metrics are the correlation, RMSE and the OLS slope of predicted on observed loss with an intercept, computed from unclipped values; predictions are clipped to the display range only when the fitted lines are drawn.

Famine outcomes and dynamic models

Event coding. Country-years are coded one throughout listed famine spells and zero otherwise, using an event list assembled from the World Peace Foundation historical catalogue and associated material[2]. The coded events vary in size and include events below 100,000 deaths, such as Malawi in 2002 and Niger in 2005. Coding is national and does not imply famine in every region of a country. Table 1 lists the coded spells for 1961–2014.

Table 1: Coded famine spells, 1961–2014.

Country	Coded years
Nigeria	1968–1970
Mauritania; Mali; Niger	1969–1974 (each country)
Ethiopia	1973–1974; 1983–1985
Uganda	1980; 2003–2006
Mozambique	1981–1985
Sudan (former)	1984–1985; 1987–1988; 1992–1994; 1998–1999; 2003–2005
Somalia	1991–1993; 2011–2012
Malawi	2002
Niger	2005
Democratic Republic of the Congo	1998–2007

Distributed lags. For Fig. 3, we estimate a joint distributed-lag regression with one lead, the current loss and three lags:

$$Y_{it} = \alpha_i + \gamma_i t + \sum_{h=-1}^{3} \beta_h L_{i,t-h} + u_{it}, \tag{6}$$

where Y is observed caloric loss or the binary famine indicator. Positive h denotes outcomes following a loss, so the one-year response corresponds to $L_{i,t-1}$. Standard errors are clustered by country. The regressions use 2,265 observations per outcome, and intervals are the coefficient ± 1.96 standard errors. Famine coefficients are multiplied by 100 to express percentage-point probability changes per percentage-point loss; the observed-loss coefficient is a ratio.

Cumulative exposure and placebos. The exposure in Fig. 5 is $K_{it} = L_{it} + L_{i,t-1}$. We estimate $F_{it} = \alpha_{i,d(t)} + \beta K_{it} + u_{it}$, absorbing country–decade fixed effects and clustering standard errors by ISO3 country. After Stata drops singletons and missing values, the estimation sample contains 2,403 country-years. The lag and lead placebo regressions replace K with $L_{i,t-2} + L_{i,t-3}$ and $L_{i,t+1} + L_{i,t+2}$, respectively.

Institutions, income and conflict

Moderator data. V-Dem v16 supplies regime type and the Liberal Democracy Index[36]. Democracy equals one for electoral and liberal democracies (regime codes 2 and 3) and zero for autocratic regimes, and we divide the Liberal Democracy Index into quartiles across all country-years. Income is V-Dem's compiled GDP per capita, which draws on the historical estimates of Fariss and colleagues[47]. Conflict indicators come from the UCDP/PRIO Armed Conflict Dataset v25.1[48]. We expand multiple country-location codes, restrict to the African sample and take the highest conflict intensity in each country-year. Minor conflict is 25–999 battle-related deaths and war is at least 1,000, following the dataset's conflict-year classification; country-years without a conflict record are coded no conflict. Landlockedness is assigned from country codes, with Ethiopia changing status in 1993. The sub-Saharan subsample excludes Algeria, Egypt, Libya, Morocco, Tunisia and South Africa.

Interaction models. We estimate heterogeneity with pooled interaction models. For a binary moderator D,

$$F_{it} = \alpha_{i,d(t)} + \beta K_{it} + \theta K_{it} D_{it} + u_{it}. \tag{7}$$

The subgroup slopes are β and $\beta + \theta$, with the uncertainty of the latter obtained from the full covariance matrix as a linear combination, and the test of heterogeneity is $H_0 : \theta = 0$. Categorical moderators enter with a loss interaction for each non-reference category. Baseline famine probability is common across moderator states within a country–decade cell. For subgroups with no or a single famine observation (middle/high income and democracy), the estimated slope is approximately zero and no confidence interval is reported.

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